



Figure 4: Paho sample app

```
mosquitto_sub -t hello
mosquitto_pub -t hello -l #payload from stdin, line wise
```

3. In the CONNECT packet, find out the protocol name, version, connect flags, keep alive value and client ID as part of the variable header. Observe the protocol name and version for both v3.1 and v3.1.1 by specifying `-V mqttv311` or `-V mqttv311` for Mosquitto clients.
4. Identify connect flags (user name present, password present, will retain, will QoS, will flag, clean session, etc) in the variable header of the CONNECT packet.
5. For each publisher, you can find the flow of CONNECT, CONNACK, PUBLISH and DISCONNECT when the client terminates. You can also identify QoS level in header flags and there will be no acknowledgment for QoS 0:

```
mosquitto_pub -t hello -m abcd
mosquitto_pub -t hello -l #hit ctrl+D to quit
```

6. For QoS 1, observe the flow of PUBLISH, PUBACK. Also match the message identifier in PUBLISH, PUBACK packets.

```
mosquitto_pub -t hello -m abcd -q 1
```

7. For QoS 2, observe the flow of PUBLISH, PUBREC, PUBREL, PUBCOMP with the same message identifier.

```
mosquitto_pub -t hello -m abcd -q 2
```

8. Try various combinations of QoS levels between *publish*, *subscribe* and observe the downgrading of QoS levels, as well as the packet flow between broker and subscribers (better to force v3.1.1 using `-V mqtt311` or use Paho clients for this testing).

```
mosquitto_sub -t hello --qos 2
mosquitto_pub -t hello -m abcd --qos 1
```

9. Observe the retain value in the header flags of the

PUBLISH message:

```
mosquitto_pub -t hello -m abcd -r
```

10. Observe the clean session in connect flags of the CONNECT packet of the subscriber, message ID as *sub.1*:

```
mosquitto_sub -t hello -c -i sub.1
```

11. Observe the flow of PINGREQ, PINGRESP for every 10 seconds if there is no message flow between client and broker for the *keep alive* duration.

```
mosquitto_sub -t hello -k 10
```

12. Identify the LWT fields with the following commands:

```
mosquitto_sub -t hello --will-topic "death" --will-payload "I am leaving..."
mosquitto_pub -t hello -l --will-topic "death" --will-payload "I am leaving..."
```

Any other client subscribing to the 'death' topic receives the message 'I am leaving...' when the above client terminates abnormally without DISCONNECT. You can try this out by terminating the client with `Ctrl+c`.

You can also try the options `--will-qos`, `--will-retain` and repeat the same steps.

13. Use the `-u` and `-P` options for authentication and observe the connect flags, username and password in the payload:

```
mosquitto_pub -u admin -P ***** -t hello -m abcd
```

The broker is assumed on the localhost in the above examples. Use the `-h` option to connect the broker on the other node.

The link <https://www.wireshark.org/docs/dfref/m/mqtt.html> summarises Wireshark support for MQTT.

This is a gist of the protocol and some implementation examples. Do refer to the given links and explore more about the protocol, its security aspects, message persistence and detailed usage of specified client libraries, tools and cloud platforms, based on these initial pointers. 

References

- [1] mqtt-v3.1.1-os.pdf, official specs from <http://mqtt.org>
- [2] <https://mosquitto.org/>
- [3] <http://www.eclipse.org/paho/>

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